

IMPORTING RELEVANT LIBRARIES

```
Insert code cell below (Ctrl+M B)
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, precision_score, recall_score, accuracy_score
```

LOAD AND EXPLORE THE DATASET

```
#Loading the dataset
df = pd.read_csv('/content/drive/MyDrive/3MTT Assessment Folder/Logistic Regression - Mini project/Invistico_Airline.csv')
```

df.head()

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	...	Online support	Ease of Online booking	s
0	satisfied	Loyal Customer	65	Personal Travel	Eco	265	0		0	0	2	...	2	3
1	satisfied	Loyal Customer	47	Personal Travel	Business	2464	0		0	0	3	...	2	3
2	satisfied	Loyal Customer	15	Personal Travel	Eco	2138	0		0	0	3	...	2	2
3	satisfied	Loyal Customer	60	Personal Travel	Eco	623	0		0	0	3	...	3	1
4	satisfied	Loyal Customer	70	Personal Travel	Eco	354	0		0	0	3	...	4	2

5 rows × 22 columns

The dataset contains airline passenger survey responses and travel information. The objective of this analysis is to predict whether a passenger is satisfied or dissatisfied based on demographic characteristics, travel details, and service ratings.

The inspection shows a mixture of categorical and numerical variables, which will require preprocessing before building a Logistic Regression model.

```
df.shape
(129880, 22)
```

This tells us the shape or size of the dataset, and they are 129880 data entries (rows) and 22 features (columns)

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 129880 entries, 0 to 129879
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   satisfaction                          129880 non-null object
1   Customer Type                        129880 non-null object
2   Age                                  129880 non-null int64
3   Type of Travel                       129880 non-null object
4   Class                                129880 non-null object
5   Flight Distance                      129880 non-null int64
6   Seat comfort                         129880 non-null int64
7   Departure/Arrival time convenient    129880 non-null int64
```

```

8 Food and drink 129880 non-null int64
9 Gate location 129880 non-null int64
10 Inflight wifi service 129880 non-null int64
11 Inflight entertainment 129880 non-null int64
12 Online support 129880 non-null int64
13 Ease of Online booking 129880 non-null int64
Insert code cell below (Ctrl+M B)
15 Leg room service 129880 non-null int64
16 Baggage handling 129880 non-null int64
17 Checkin service 129880 non-null int64
18 Cleanliness 129880 non-null int64
19 Online boarding 129880 non-null int64
20 Departure Delay in Minutes 129880 non-null int64
21 Arrival Delay in Minutes 129487 non-null float64
dtypes: float64(1), int64(17), object(4)
memory usage: 21.8+ MB

```

The dataset consists of both numerical and categorical features. Variables such as Customer Type, Type of Travel, and Class are stored as categorical values and will need to be encoded into numerical representations before modeling.

```
df.duplicated().any()
```

```
np.False_
```

This means there are no duplicates in the dataset

```
df.isnull().sum()
```

	0
satisfaction	0
Customer Type	0
Age	0
Type of Travel	0
Class	0
Flight Distance	0
Seat comfort	0
Departure/Arrival time convenient	0
Food and drink	0
Gate location	0
Inflight wifi service	0
Inflight entertainment	0
Online support	0
Ease of Online booking	0
On-board service	0
Leg room service	0
Baggage handling	0
Checkin service	0
Cleanliness	0
Online boarding	0
Departure Delay in Minutes	0
Arrival Delay in Minutes	393

```
dtype: int64
```

There are 393 missing values, only in the `Arrival Delay in Minutes` column though. The below code is to further investigate if there is any logical reason why the values are omitted (or if there are any systemic reasons for omission)

```
df[df.isnull().any(axis=1)]
```

	satisfaction	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	...	Online support	Ease of Online booking
Insert code cell below (Ctrl+M B)		Loyal Customer	21	Personal Travel	Eco	4804	0		2	0	4	...	5
246	satisfied	Loyal Customer	28	Personal Travel	Eco	1364	0		3	0	4	...	4
710	satisfied	Loyal Customer	28	Personal Travel	Eco	1792	0		5	0	3	...	5
736	satisfied	Loyal Customer	9	Personal Travel	Eco	1862	0		5	0	4	...	3
818	dissatisfied	Loyal Customer	9	Personal Travel	Eco	1668	1		0	1	1	...	4
...	...	...	...	...	...	...	...		...	...	...	...	...
127739	satisfied	Loyal Customer	44	Business travel	Business	3431	2		2	3	2	...	4
128131	satisfied	Loyal Customer	39	Business travel	Business	1595	4		4	4	4	...	5
128710	satisfied	Loyal Customer	45	Business travel	Eco	871	5		1	1	1	...	2
129774	satisfied	disloyal Customer	49	Personal Travel	Eco	1594	5		4	5	2	...	3
129873	satisfied	disloyal Customer	63	Personal Travel	Eco	1942	5		5	4	4	...	3

393 rows × 22 columns

df.describe()

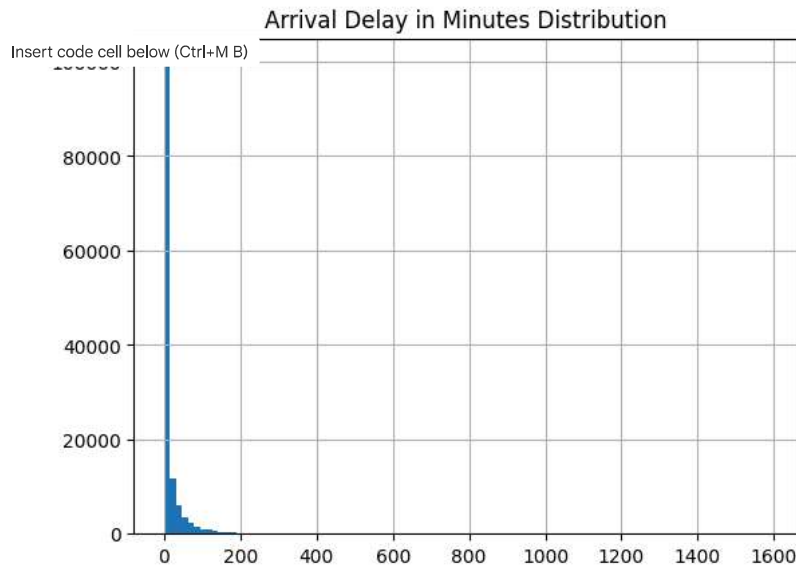
	Age	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	Inflight wifi service	Inflight entertainment
count	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000	129880.000000
mean	39.427957	1981.409055	2.838597	2.990645	2.851994	2.990422	3.249130	3.383477
std	15.119360	1027.115606	1.392983	1.527224	1.443729	1.305970	1.318818	1.346059
min	7.000000	50.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	27.000000	1359.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000
50%	40.000000	1925.000000	3.000000	3.000000	3.000000	3.000000	3.000000	4.000000
75%	51.000000	2544.000000	4.000000	4.000000	4.000000	4.000000	4.000000	4.000000
max	85.000000	6951.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000

Summary Statistics Interpretation

- **Age:** Passengers range from 7 to 85 years old, with average age approximately 39 years.
- **Flight Distance:** Typical flights are medium-haul (median ~1925 km), but some extend up to ~6951 km.
- **Service Ratings:** Most service-related features (seat comfort, food, entertainment, cleanliness, etc.) are rated on a 0–5 scale. Median values cluster around 3–4, suggesting average to good satisfaction.
- **Delays:** Departure and arrival delays are highly skewed. While the median delay is 0 minutes (most flights on time), extreme outliers exist (up to ~1592 minutes).
- **Key Insight:** The dataset shows generally average service ratings, with delays being the most variable factor. This variability in delays may strongly influence customer satisfaction.

✖ HANDLE MISSING VALUES

```
df['Arrival Delay in Minutes'].hist(bins=100)
plt.title("Arrival Delay in Minutes Distribution")
plt.show()
```



Arrival Delay Distribution Interpretation

The histogram reveals a highly **right-skewed distribution** of arrival delays.

- Most flights arrive on time or within a few minutes of schedule.
- A small number of flights experience extreme delays, creating a long tail on the right.
- This pattern suggests that while delays are generally rare, they can be severe when they occur.
- The skewness justifies using the **median** rather than the mean for imputation.

```
# Here we impute Arrival Delay in Minutes column with the median
```

```
median_delay = df['Arrival Delay in Minutes'].median()
df['Arrival Delay in Minutes'].fillna(median_delay, inplace=True)
```

/tmp/ipykernel_2373/4128011878.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chain. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df

```
df['Arrival Delay in Minutes'].fillna(median_delay, inplace=True)
```

```
#Confirming if it is effected
```

```
df['Arrival Delay in Minutes'].isnull().sum()
```

```
np.int64(0)
```

FEATURE ENGINEERING

Several predictors in the dataset are categorical and cannot be used directly by Scikit-learn.

These variables were encoded into numerical representations so that the Logistic Regression algorithm could process them effectively.

Examples of these variables include: `satisfaction`, `Customer Type`, `Type of Travel`, and `Travel Class` variables

```
df['satisfaction'].value_counts()
```

```

count
satisfaction
satisfied    71087
dtype: int64

```

Insert code cell below (Ctrl+M B)

The **satisfaction** column represents the target variable for prediction.

- Out of 129,880 passengers, **71,087 (≈55%)** reported being *satisfied*, while **58,793 (≈45%)** were *dissatisfied*.
- This indicates a **moderately balanced dataset**, meaning the model will not be heavily biased toward one class.
- The slight majority of satisfied customers suggests overall positive experiences, but there's still a significant portion of dissatisfied passengers — providing valuable insights for improvement.

```
df['Customer Type'].value_counts()
```

```

count
Customer Type
Loyal Customer    106100
disloyal Customer  23780
dtype: int64

```

```
df['Type of Travel'].value_counts()
```

```

count
Type of Travel
Business travel  89693
Personal Travel  40187
dtype: int64

```

```
df['Class'].value_counts()
```

```

count
Class
Business  62160
Eco       58309
Eco Plus  9411
dtype: int64

```

✓ ENCODE CATEGORICAL VARIABLES

```

le = LabelEncoder()
for col in ['satisfaction', 'Customer Type', 'Type of Travel', 'Class']:
    df[col] = le.fit_transform(df[col])

```

Logistic Regression requires a binary target variable. Therefore, the satisfaction variable was encoded such that:

1 = Satisfied 0 = Dissatisfied

also Customer Type along with the other two categorical data was converted into numerical forms so that it could be interpreted by the model.

Confirming if the Encoding Process worked

```
df['satisfaction'].value_counts()
```

	count
satisfaction	
0	58793

dtype: int64

```
df['Customer Type'].value_counts()
```

	count
Customer Type	
0	106100
1	23780

dtype: int64

```
df['Type of Travel'].value_counts()
```

	count
Type of Travel	
0	89693
1	40187

dtype: int64

```
df['Class'].value_counts()
```

	count
Class	
0	62160
1	58309
2	9411

dtype: int64

✓ SPLIT DATA FOR TRAINING AND TESTING

```
X = df.drop('satisfaction', axis=1)
y = df['satisfaction']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

The dataset was divided into training and testing subsets using an 80/20 split.

```
X.head()
```

	Customer Type	Age	Type of Travel	Class	Flight Distance	Seat comfort	Departure/Arrival time convenient	Food and drink	Gate location	Inflight wifi service	...	Online support	Ease of Online booking	On-board service
Insert code cell below (Ctrl+M B)				1	1	265	0	0	2	2	...	2	3	3
1	0	47	1	0	2464	0	0	0	3	0	...	2	3	4
2	0	15	1	1	2138	0	0	0	3	2	...	2	2	3
3	0	60	1	1	623	0	0	0	3	3	...	3	1	1
4	0	70	1	1	354	0	0	0	3	4	...	4	2	2

5 rows × 15 columns

```
y.head()
```

```

satisfaction
0      1
1      1
2      1
3      1
4      1

```

```
dtype: int64
```

▼ BUILD AND FIT LOGISTIC REGRESSION MODEL

```

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

```

```

/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```

  ▼ LogisticRegression ⓘ ?
  LogisticRegression(max_iter=1000)

```

A Logistic Regression model was trained to estimate the probability that a passenger is satisfied.

Unlike Linear Regression, Logistic Regression is specifically designed for binary classification problems and produces probabilities between 0 and 1.

▼ MODEL EVALUATION

```

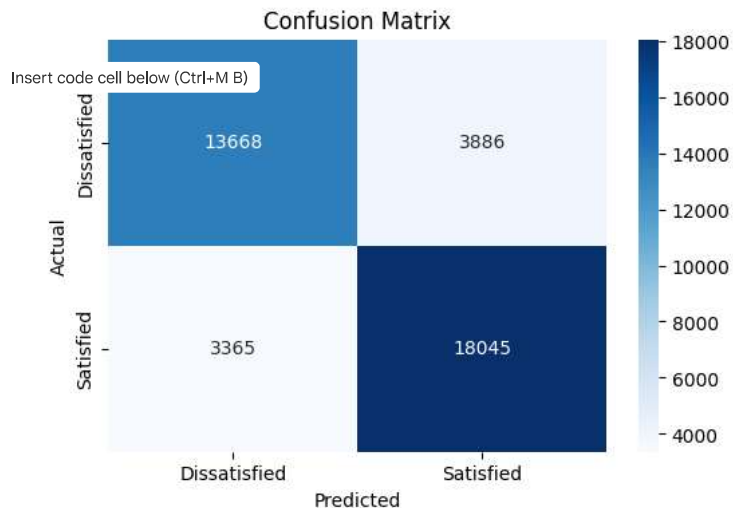
# Generate predictions
y_pred = model.predict(X_test)

# Create confusion matrix
cm = confusion_matrix(y_test, y_pred)

# Plot with seaborn
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Dissatisfied', 'Satisfied'],
            yticklabels=['Dissatisfied', 'Satisfied'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')

```

```
plt.show()
```



The confusion matrix shows that the model correctly identified 13,668 dissatisfied passengers and 18,045 satisfied passengers.

However, 3,365 dissatisfied passengers were incorrectly classified as satisfied while 3,886 satisfied passengers were incorrectly classified as dissatisfied.

Overall, the model demonstrates strong classification performance with relatively balanced error rates across both classes.

```
recall = recall_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
```

```
Accuracy: 0.8139051432091161
Precision: 0.8228078974966941
Recall: 0.842830453059318
```

The model achieved an Accuracy of 81.4%, Precision of 82.3% and a Recall rate of 84.3%.

Precision indicates that when the model predicts a passenger is satisfied, it is correct approximately 82% of the time.

Recall indicates that the model successfully identifies approximately 84% of all truly satisfied passengers.

Together, these metrics suggest that the model performs well in identifying satisfied customers while maintaining a relatively low false-positive rate.

```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_[0]
}).sort_values(by='Coefficient', ascending=False)
```

```
coefficients
```


	Feature	Coefficient
10	Inflight entertainment	0.610292
5	Seat comfort	0.303425
Insert code cell below (Ctrl+M B)		
	Online booking	0.288175
13	On-board service	0.278618
16	Checkin service	0.227864
14	Leg room service	0.175048
18	Online boarding	0.085787
11	Online support	0.041933
15	Baggage handling	0.039973
8	Gate location	0.026125
17	Cleanliness	0.004735
19	Departure Delay in Minutes	0.001644
4	Flight Distance	-0.000363
20	Arrival Delay in Minutes	-0.006338
1	Age	-0.024832
9	Inflight wifi service	-0.151486
7	Food and drink	-0.191809
6	Departure/Arrival time convenient	-0.281364
2	Type of Travel	-0.616048
3	Class	-0.793179
0	Customer Type	-2.520130

Key Positive Drivers

Inflight Entertainment (+0.610)

Inflight entertainment is the strongest positive predictor of satisfaction in the model.

Passengers who give higher ratings for inflight entertainment are substantially more likely to be satisfied compared with passengers who provide lower ratings.

Seat Comfort (+0.303)

Seat comfort is the second strongest positive driver of satisfaction, suggesting that passenger comfort plays a critical role in the overall travel experience.

Ease of Online Booking (+0.288)

A smoother booking experience increases the likelihood of customer satisfaction. This highlights the importance of digital convenience throughout the customer journey.

On-board Service (+0.279)

Passengers who perceive better service from airline staff are more likely to report satisfaction.

Check-in Service (+0.228)

Efficient and pleasant check-in experiences contribute positively to overall customer satisfaction.

Key Negative Drivers

Customer Type (-2.520)

Customer Type has the strongest negative coefficient in the model.

This suggests that one customer segment is substantially less likely to be satisfied than the reference group.

Class (-0.793)

Travel class significantly influences satisfaction outcomes. The negative coefficient indicates that some classes may be associated with lower satisfaction levels relative to the reference category.

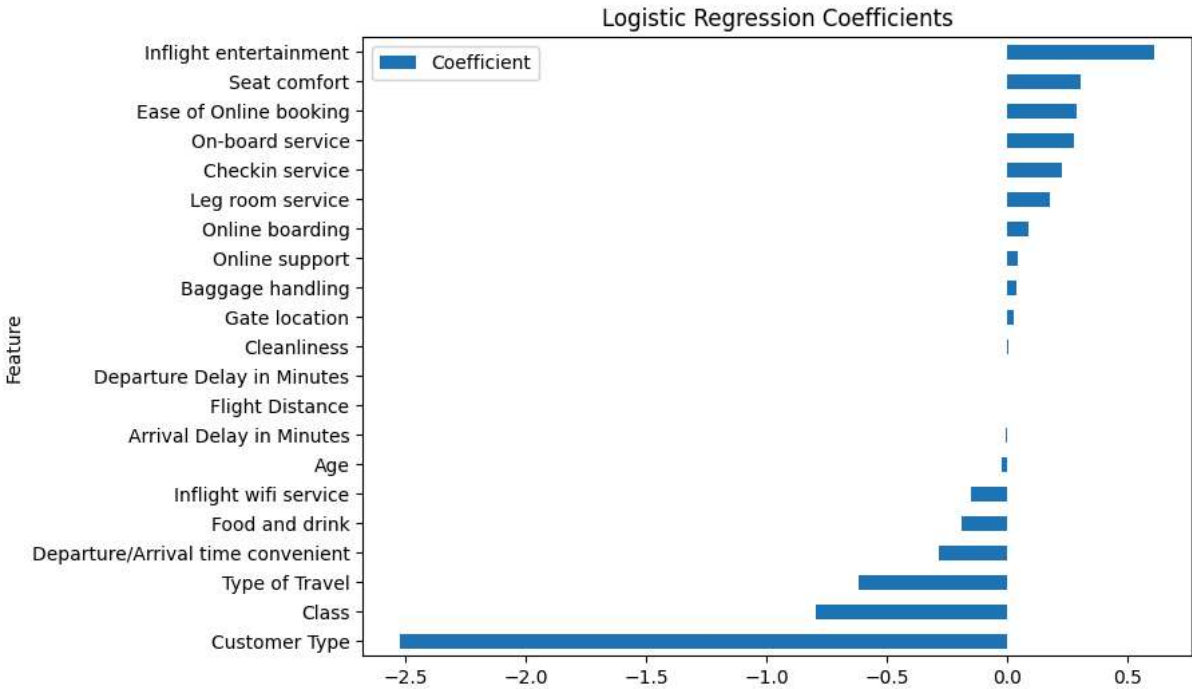
Type of Travel (-0.616)

Passenger travel purpose appears to influence satisfaction, indicating that business and personal travelers may have different expectations and experiences.

Departure/Arrival Time Convenience (-0.281)

Passengers who perceive flight schedules as inconvenient are less likely to report satisfaction.

```
# Grapical representation of these coefficient
coefficients.sort_values('Coefficient').plot(
    x='Feature',
    y='Coefficient',
    kind='barh',
    figsize=(8,6)
)
plt.title("Logistic Regression Coefficients")
plt.show()
```



✦ BUSINESS INSIGHTS

The analysis suggests that customer satisfaction is driven primarily by service quality and travel experience rather than operational metrics.

Features such as inflight entertainment, seat comfort, online booking, onboard service, and check-in service demonstrate stronger effects than delay-related variables.

Interestingly, arrival and departure delays have relatively small coefficients, suggesting that passengers may place greater emphasis on the overall travel experience than minor scheduling disruptions.

✦ BUSINESS RECOMMENDATIONS

Based on the Logistic Regression model, the airline should prioritize investments in customer experience initiatives that are most strongly associated with satisfaction.

Recommendation 1: Improve Inflight Entertainment

Inflight entertainment produced the largest positive coefficient in the model (+0.610), making it the strongest driver of satisfaction. Expanding entertainment options, improving content quality, and ensuring system reliability could increase passenger satisfaction levels.

Recommendation 2: Enhance Seat Comfort

Seat comfort emerged as the second strongest positive factor. Investments in seating quality, legroom optimization, and cabin ergonomics could improve the passenger experience.

Recommendation 3: Streamline Digital Services

Ease of online booking and online boarding both contributed positively to satisfaction. The airline should continue investing in user-friendly digital platforms and self-service technologies.

Recommendation 4: Strengthen Customer Service Training

On-board service and check-in service were among the strongest positive predictors. Training frontline employees to provide consistent, high-quality service could yield measurable improvements in customer satisfaction.

Recommendation 5: Target At-Risk Customer Segments

Customer Type and Type of Travel showed strong negative effects. Further investigation should identify which passenger groups are less satisfied so that targeted retention and loyalty strategies can be developed.

Start coding or generate with AI.

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.